INDEFINITE THETA HESSIANS AT GREEN-FUNCTION PARAMETERS

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Abstract. We construct triply connected circle domains for which the logarithmic theta Hessian at the Green-function parameter is indefinite, disproving the negative-definiteness conjecture of Saitoh in Yamada's theta-function formulation.

1. Introduction

Let Ω be a bounded planar domain with $g+1$ analytic boundary components and Schottky double $\widehat{\Omega}$. Following Yamada [Yam99], we choose a symmetric canonical homology basis $A_1,\dots,A_g,B_1,\dots,B_g$ on $\widehat{\Omega}$, with the B_j represented by the inner boundary components of Ω. We normalize the holomorphic differentials $\eta_1,\dots,\eta_g$ by

$$\int_{A_j}\eta_k = 2\pi i\delta_{jk}, \qquad \tau_{jk}=\int_{B_j}\eta_k.$$

The orientations and construction of this basis are specified in Section 3. The period matrix $\tau=(\tau_{jk})$ is real symmetric and negative definite, and the theta function is

$$\theta_\tau(z)=\sum_{n\in\mathbb{Z}^g}\exp\left(\frac{1}{2}n^T\tau n+n^Tz\right), \qquad z\in\mathbb{C}^g. \tag{1}$$

We refer to Fay [Fay73] for background. On $i\mathbb{R}^g/(2\pi i\mathbb{Z}^g)$, the function θ_τ is strictly positive by Poisson summation; see (18) for $g=2$, whose derivation applies in any dimension. Wherever $\theta_\tau(e)\neq 0$, define the *holomorphic* Hessian of $\log\theta_\tau$ by

$$H_\tau(e)=\left(\frac{\partial^2\log\theta_\tau}{\partial z_j\partial z_k}(e)\right)_{j,k=1}^{g}. \tag{2}$$

For $e=it\in i\mathbb{R}^g$, this matrix is well-defined and real symmetric, and satisfies

$$H_\tau(it)=-\nabla_t^2\Phi_\tau(t), \qquad \Phi_\tau(t)=\log\theta_\tau(it). \tag{3}$$

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Let $G_\Omega(z,p)$ be the Green function that vanishes on $\partial\Omega$ and has singularity $\log|z-p|$. The *Green differential* is

$$\mathcal{G}_p(z) = 2\partial G_\Omega(z,p) = 2\frac{\partial G_\Omega}{\partial z}(z,p)\,dz.$$

It is holomorphic on $\Omega\setminus\{p\}$ and has a simple pole of residue 1 at p. Let D_p be its degree-g zero divisor in Ω, counted with multiplicities. Set

$$e_0(p) = D_p - p - \Delta, \tag{4}$$

where Δ is the Riemann divisor class of degree $g-1$, with $2\Delta = K_{\widehat{\Omega}}$, in the convention of [Fay73, Yam99]. The degree-zero class in (4) is identified with a point of the Jacobian by the Abel–Jacobi map. It has a representative in $i\mathbb{R}^g$, and $H_\tau(e_0(p))$ is independent of the period ambiguity. Equivalently, by Theorem 5.1, $e_0(p)$ is the unique maximizer in (53).

Saitoh [Sai79, pp. 276–277, equation (7.11)] proved that, on a doubly connected domain, a product of two Rudin kernels dominates the Bergman kernel in the order of positive definite kernels, and asked whether this extends to arbitrary compact bordered Riemann surfaces. This comparison is stronger than its diagonal consequence, equation (7.12) of that paper, and differs from the earlier comparison with the *exact* Bergman kernel [Sai77, Corollary 3.1].

In the theta-function formulation, the proposed conclusion is

$$H_\tau(e_0(p)) < 0. \tag{5}$$

Yamada [Yam99, p. 1399] attributes this problem to Saitoh, citing [Sai88, p. 37]; we follow Yamada's attribution. See also [Yam98, p. 41]. We call (5) *Saitoh's negative-definiteness conjecture in Yamada's theta-function formulation*. The matrix conjecture requires negativity in every nonzero real direction.

Guan [Gua19] proved the scalar comparison between the conjugate Hardy kernel and the Bergman kernel. Further generalizations were obtained by Guan–Yuan [GY24, GY22, GY23, GY26] and Guan–Li–Yuan [GLY26]. Section 6 explains how Guan's scalar inequality corresponds to negativity of $H_\tau(e_0(p))$ in a distinguished direction.

For $g=0$, the kernel comparison is an equality, $\theta\equiv 1$, and the Hessian is an empty matrix. For $g=1$, Saitoh's doubly connected result gives $H_\tau(e_0(p))<0$ for every $p\in\Omega$ [Sai79, §7, equations (7.1), (7.11)–(7.12)].

We disprove (5) in genus $g=2$, the smallest possible genus and the first admitting a transverse direction.

Fix $0<\rho<1$ and put

$$M_\rho = \log(1/\rho), \qquad d_* = \frac{2\pi^2}{M_\rho}. \tag{6}$$

For small $\varepsilon > 0$, let $c = 1 - \varepsilon$ and define

$$K_{\pm}(c) = \left\{ z \in \mathbb{D} : \left| \frac{z \mp c}{1 \mp cz} \right| \leq \rho \right\}, \tag{7}$$

$$\Omega_\varepsilon = \mathbb{D} \setminus \big(K_+(c) \cup K_-(c) \big), \qquad p_\varepsilon = i(1 - \varepsilon^5). \tag{8}$$

For $0 < \varepsilon < 1 - \rho$, these are disjoint closed Euclidean disks strictly inside $\mathbb{D}$, so Ω_ε is a regular triply connected domain. The centers and the common radius of the disks are, respectively,

$$\pm \frac{c(1-\rho^2)}{1 - c^2\rho^2}, \qquad \frac{\rho(1-c^2)}{1-c^2\rho^2}. \tag{9}$$

The positive center in (9) exceeds the radius exactly when $c > \rho$, since

$$c(1-\rho^2) - \rho(1-c^2) = (c-\rho)(1+c\rho) > 0.$$

Neither disk meets the imaginary axis, and $p_\varepsilon \in \Omega_\varepsilon$. Figure 1 illustrates the geometry of this family.

Let τ_ε be the period matrix of $\widehat{\Omega}_\varepsilon$. Sections 3 and 4 establish

$$\tau_\varepsilon = -\pi C_\varepsilon = \begin{pmatrix} -d_\varepsilon & b_\varepsilon \\ b_\varepsilon & -d_\varepsilon \end{pmatrix},$$

$$(C_\varepsilon)_{jk} = \int_{\Omega_\varepsilon} \nabla\omega_j \cdot \nabla\omega_k \, dA, \qquad j,k \in \{+,-\},$$

where $\omega_\pm$ are the harmonic measures of the holes. Thus d_ε and b_ε are determined by the domain.

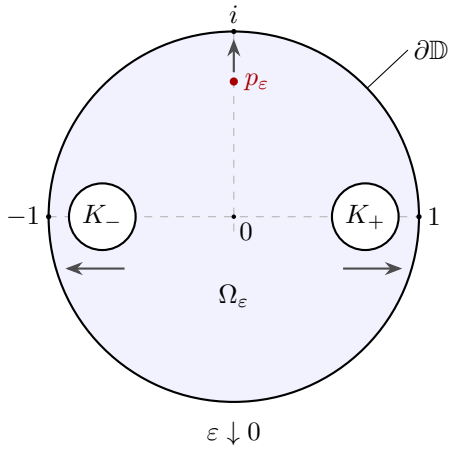


FIGURE 1. The shaded domain Ω_ε is the unit disk with the two closed disks $K_\pm(c)$ removed. As $\varepsilon \downarrow 0$, the holes shrink towards ± 1 and the pole $p_\varepsilon = i(1-\varepsilon^5)$ tends to i.

For $d > 0$, write

$$P_d(t) = \sum_{n\in\mathbb{Z}} e^{-dn^2/2+int}, \qquad \ell_d(t) = \log P_d(t). \tag{10}$$

Lemma 2.1 proves that there is a unique $t_d \in (\pi/2, \pi)$ such that $\ell_d''(t_d) = 0$, and that

$$\kappa_d = \ell_d'''(t_d) > 0. \tag{11}$$

Let $\kappa_* := \kappa_{d_*}$.

Theorem 1.1. *Fix $0 < \rho < 1$ and let $\Omega_\varepsilon, p_\varepsilon$ be given by (8). In the normalized basis associated with the two inner boundary components, the matrix*

$$H_{\tau_\varepsilon}\big(e_0(p_\varepsilon)\big)$$

has eigenvectors $(1, 1)$ and $(1, -1)$. Their respective eigenvalues satisfy

$$\lambda_{\parallel}(\varepsilon) = -\frac{\pi\kappa_*}{M_\rho}\varepsilon^6 + O(\varepsilon^7), \tag{12}$$

$$\lambda_{\perp}(\varepsilon) = \frac{\pi^4\kappa_*^2}{M_\rho^4}\varepsilon^4 + O(\varepsilon^5). \tag{13}$$

For all sufficiently small positive ε, the matrix has one strictly positive and one strictly negative eigenvalue.

For every sufficiently small $\varepsilon > 0$, the double is a smooth genus-two surface.

As the coupling b tends to zero, the theta function approaches a product of elliptic theta functions. At an auxiliary extremum, one Hessian eigenvalue vanishes and the other equals $b^2\kappa^2 + O(b^3) > 0$. The Abel displacement δ moves the extremum, making the first eigenvalue negative at order δ. Our geometry ensures $\delta = o(b^2)$, so the second eigenvalue remains positive.

Section 2 gives the theta expansion. Sections 3 and 4 derive the period and harmonic-measure estimates. Section 5 identifies the extremum with the Green-function parameter and completes the proof.

2. A perturbation of a product theta function

2.1. A nondegenerate one-dimensional maximum.

Lemma 2.1. *For every $d > 0$, the function $-\ell_d'$ on $\mathbb{R}/(2\pi\mathbb{Z})$ has a unique maximum at a point $t_d \in (\pi/2, \pi)$. This maximum is nondegenerate. More precisely, $\ell_d''(t_d) = 0$ and $\kappa_d > 0$. The functions $d \mapsto t_d$ and $d \mapsto \kappa_d$ are smooth.*

Proof. Set $h_d = -\ell_d''$ and $q = e^{-d/2}$. The Jacobi product formula is

$$P_d(t) = \prod_{m\geq 1}(1 - q^{2m})(1 + 2q^{2m-1}\cos t + q^{4m-2}).$$

Every factor is positive for real t. For $f_r(t) = 1 + 2r\cos t + r^2$ one has

$$-\frac{d^2}{dt^2}\log f_r(t) = \frac{2r\cos t}{f_r(t)} + \frac{4r^2\sin^2 t}{f_r(t)^2} = \frac{2r((1+r^2)\cos t + 2r)}{f_r(t)^2}.$$

With $r_m = q^{2m-1}$, summing these logarithmic derivatives gives

$$h_d(t) = 2\sum_{m\geq 1} \frac{r_m((1+r_m^2)\cos t + 2r_m)}{(1+2r_m\cos t + r_m^2)^2}. \tag{14}$$

The product and its real derivatives converge uniformly. In particular,

$$h_d(0) = 2\sum_{m\geq 1}\frac{r_m}{(1+r_m)^2} > 0,$$

$$h_d(\pi) = -2\sum_{m\geq 1}\frac{r_m}{(1-r_m)^2} < 0,$$

$$h_d(\pi/2) = 4\sum_{m\geq 1}\frac{r_m^2}{(1+r_m^2)^2} > 0.$$

To prove strict monotonicity on $(0,\pi)$, note that P_d has simple zeros at

$$\pi + id/2 + 2\pi\mathbb{Z} + id\mathbb{Z},$$

and satisfies $P_d(t+id) = e^{d/2-it}P_d(t)$. Completing the square in the defining series and replacing $n+1$ by a new index gives

$$\sum_{n\in\mathbb{Z}} e^{-dn^2/2 - dn + int} = e^{d/2-it}\sum_{n\in\mathbb{Z}} e^{-d(n+1)^2/2 + i(n+1)t}.$$

Thus h_d extends to an elliptic function with periods $2\pi, id$ and one double pole, of principal part $(t-\pi-id/2)^{-2}$, per fundamental parallelogram. Subtracting the corresponding Weierstrass function leaves a constant:

$$h_d(t) = \wp(t-\pi-id/2;\, 2\pi\mathbb{Z} + id\mathbb{Z}) + C_d. \tag{15}$$

See also [DLMF, §23.6]. Since $\wp'$ vanishes only at the nonzero half-periods, h_d' has no zero on $(0,\pi)$. The endpoint values therefore imply $h_d' < 0$ there.

Hence h_d has a unique zero $t_d \in (\pi/2, \pi)$. By evenness and periodicity, its only other zero is $2\pi - t_d$. These are respectively the unique maximum and minimum of $-\ell_d'$, and $\kappa_d = -h_d'(t_d) > 0$. The implicit function theorem gives smooth dependence on d. □

2.2. **Uniform analytic expansion in the coupling.** For $|b| < d$ define

$$\Theta_{d,b}(z_1,z_2) = \sum_{m,n\in\mathbb{Z}} e^{-\frac{d}{2}(m^2+n^2)+bmn+mz_1+nz_2}, \tag{16}$$

$$\Phi_{d,b}(t_1,t_2) = \log\Theta_{d,b}(it_1,it_2), \qquad \mathcal{H}_{d,b}(it) = -\nabla_t^2\Phi_{d,b}(t). \tag{17}$$

All variables t_j are understood modulo 2π. Set

$$A = \begin{pmatrix} d & -b \\ -b & d \end{pmatrix} > 0.$$

With the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^2} f(x) e^{-ix\cdot\xi}\, dx, \qquad f_t(x) = e^{-x^T Ax/2 + it\cdot x},$$

the Gaussian transform is

$$\widehat{f}_t(\xi) = \frac{2\pi}{\sqrt{\det A}} \exp\left(-\frac{1}{2}(\xi - t)^T A^{-1}(\xi - t)\right).$$

Poisson summation gives

$$\sum_{n\in\mathbb{Z}^2} f_t(n) = \sum_{k\in\mathbb{Z}^2} \widehat{f}_t(2\pi k).$$

Both sides converge absolutely, and hence

$$\Theta_{d,b}(it) = \frac{2\pi}{\sqrt{\det A}} \sum_{k\in\mathbb{Z}^2} \exp\left(-\frac{1}{2}(t - 2\pi k)^T A^{-1}(t - 2\pi k)\right) > 0. \tag{18}$$

The real logarithm in (17) is defined everywhere on the real torus.

Throughout this section, d lies in a small compact neighborhood $\mathcal{I} \Subset (0, \infty)$ of $d_* > 0$, and all remainder estimates are uniform in d.

Lemma 2.2. *Put $\ell = \ell_d$, and abbreviate $p_j = \ell'(t_j)$, $q_j = \ell''(t_j)$. Then, as $b \to 0$,*

$$\begin{aligned}\Phi_{d,b}(t_1, t_2) &= \ell(t_1) + \ell(t_2) - bp_1p_2 \\ &\quad + \frac{b^2}{2}\big(q_1q_2 + q_1p_2^2 + p_1^2q_2\big) + O(b^3).\end{aligned} \tag{19}$$

The expansion holds in every fixed C^r norm on the real torus, uniformly for $d \in \mathcal{I}$.

Proof. For $|b| \le b_0 < \min \mathcal{I}$, the quadratic exponent is bounded above by $-(\min \mathcal{I} - b_0)(m^2 + n^2)/2$. Differentiating a summand in b, in d, or in t only introduces polynomial factors. All such differentiated series converge uniformly by Gaussian summability. More explicitly, the scalar Taylor remainder obeys

$$\left| e^{bmn} - 1 - bmn - \frac{b^2m^2n^2}{2} \right| \le \frac{|b|^3|mn|^3}{6} e^{b_0|mn|}.$$

After multiplication by the uncoupled Gaussian, $2|mn| \le m^2 + n^2$ gives a summable majorant, even after any fixed number of derivatives. The remainder is therefore $O(b^3)$ in every fixed C^r norm.

Let $P = P_d$ and $\Pi_d(t_1, t_2) = P(t_1)P(t_2)$. Since differentiating twice, once in each real variable, multiplies a Fourier mode by $-mn$, expansion of e^{bmn} gives

$$\Theta_{d,b}(it) = \Pi_d - b\,\partial_1\partial_2\Pi_d + \frac{b^2}{2}\partial_1^2\partial_2^2\Pi_d + O(b^3).$$

Dividing by Π_d and using $P'/P = \ell'$ and $P''/P = \ell'' + (\ell')^2$, we obtain

$$\frac{\Theta_{d,b}(it)}{\Pi_d} = 1 - bp_1p_2 + \frac{b^2}{2}(q_1 + p_1^2)(q_2 + p_2^2) + O(b^3).$$

If the last display is written as $1 + bx + b^2y + O(b^3)$, then

$$\log(1 + bx + b^2y + O(b^3)) = bx + b^2\left(y - \frac{x^2}{2}\right) + O(b^3), \quad x = -p_1p_2.$$

The quadratic coefficient is

$$y - \frac{x^2}{2} = \frac{1}{2}\left[(q_1 + p_1^2)(q_2 + p_2^2) - p_1^2p_2^2\right] = \frac{1}{2}(q_1q_2 + q_1p_2^2 + p_1^2q_2).$$

This proves (19). Positivity and compactness give uniform lower bounds for Π_d and, for small b, for $\Theta_{d,b}(it)$, so the logarithmic expansion has the same C^r remainder bounds. □

2.3. **A positive transverse eigenvalue.** Let $\mathbf{1} = (1, 1)$ and $\partial_{\mathbf{1}} = \partial_1 + \partial_2$, and define

$$S_{d,b}(t) = -\partial_{\mathbf{1}}\Phi_{d,b}(t). \tag{20}$$

Proposition 2.3. *For sufficiently small $|b|$, $S_{d,b}$ has a unique global maximum on the real torus. It is nondegenerate and has the form $(\sigma_{d,b}, \sigma_{d,b})$, where $\sigma_{d,b} = t_d + O(b)$. At this point the eigenvalues of $\mathcal{H}_{d,b}$ in the directions $(1, 1)$ and $(1, -1)$ are, respectively,*

$$0, \qquad b^2\kappa_d^2 + O(b^3). \tag{21}$$

Moreover, if $\Lambda_{\parallel}(\sigma)$ is the $(1, 1)$ eigenvalue of $\mathcal{H}_{d,b}(i\sigma, i\sigma)$, then

$$\Lambda'_{\parallel}(\sigma_{d,b}) = -\kappa_d + O(b). \tag{22}$$

Proof. At $b = 0$ the function in (20) is $-\ell'_d(t_1) - \ell'_d(t_2)$. Its unique global maximum is (t_d, t_d), with Hessian $-\kappa_d I_2$. Set

$$\mathcal{F}(t, b) = \nabla_t S_{d,b}(t), \qquad t^0 = (t_d, t_d).$$

Since $\mathcal{F}(t^0, 0) = 0$ and $\nabla_t\mathcal{F}(t^0, 0) = -\kappa_d I_2$, the implicit function theorem gives a unique smooth branch of local maxima $t(b) = t^0 + O(b)$. Choose a small convex neighborhood of t^0 on which $S_{d,0}$ is strictly concave. Compactness gives a positive value gap outside this neighborhood. Both properties persist under small C^2 perturbations, proving global uniqueness, uniformly for $d \in \mathcal{I}$ after shrinking $\mathcal{I}$.

Exchange symmetry and uniqueness imply $t(b) = (\sigma, \sigma)$, where $\sigma = t_d + O(b)$.

Write $\Phi = \Phi_{d,b}$. Stationarity of $S = -\Phi_1 - \Phi_2$ gives

$$0 = \partial_1 S(\sigma, \sigma) = -(\Phi_{11} + \Phi_{12})(\sigma, \sigma), \qquad 0 = \partial_2 S(\sigma, \sigma) = -(\Phi_{21} + \Phi_{22})(\sigma, \sigma).$$

The first stationarity equation is

$$\Phi_{11}(\sigma, \sigma) + \Phi_{12}(\sigma, \sigma) = 0. \tag{23}$$

Set $p = \ell_d'(\sigma)$, $q = \ell_d''(\sigma)$ and $r = \ell_d'''(\sigma)$. Since $\sigma = t_d + O(b)$, we have $p = O(1)$, $q = O(b)$ and $r = \kappa_d + O(b)$. With $r_j = \ell_d'''(t_j)$,

$$\partial_1\partial_2(p_1p_2) = q_1q_2, \qquad \partial_1\partial_2(q_1q_2) = r_1r_2,$$
$$\partial_1\partial_2(q_1p_2^2) = 2r_1p_2q_2, \qquad \partial_1\partial_2(p_1^2q_2) = 2p_1q_1r_2.$$

Using (19), we obtain

$$\Phi_{12}(\sigma,\sigma) = -bq^2 + \frac{b^2}{2}(r^2 + 4pqr) + O(b^3) = \frac{b^2}{2}\kappa_d^2 + O(b^3). \tag{24}$$

Exchange symmetry gives, for every real s,

$$\mathcal{H}_{d,b}(is, is) = \begin{pmatrix} A_{\mathrm{diag}}(s) & A_{\mathrm{off}}(s) \\ A_{\mathrm{off}}(s) & A_{\mathrm{diag}}(s) \end{pmatrix}.$$

Here $A_{\mathrm{diag}}(s) = -\Phi_{11}(s,s)$ and $A_{\mathrm{off}}(s) = -\Phi_{12}(s,s)$. The eigenvectors are $(1,1)$ and $(1,-1)$, with eigenvalues $A_{\mathrm{diag}} + A_{\mathrm{off}}$ and $A_{\mathrm{diag}} - A_{\mathrm{off}}$. At $s = \sigma$, (23) gives $A_{\mathrm{diag}} + A_{\mathrm{off}} = 0$, so the transverse eigenvalue is $2\Phi_{12}$, proving (21). Finally,

$$\Lambda_{\parallel}(\sigma) = -(\Phi_{11} + \Phi_{12})(\sigma,\sigma).$$

At $b = 0$, its derivative at t_d is $-\kappa_d$. Smooth dependence proves (22). □

2.4. **Moving the maximum into a finite theta ratio.** For $\delta > 0$ set

$$R_{d,b,\delta}(t) = \Phi_{d,b}(t - \delta\mathbf{1}) - \Phi_{d,b}(t). \tag{25}$$

Proposition 2.4. *For $d \in \mathcal{I}$ and sufficiently small $|b|, \delta > 0$, (25) has a unique global maximum, at a point $v_{d,b,\delta}\mathbf{1}$. It satisfies*

$$v_{d,b,\delta} = \sigma_{d,b} + \frac{\delta}{2} + O(\delta^2). \tag{26}$$

At this maximum the two eigenvalues of $\mathcal{H}_{d,b}$ satisfy

$$\lambda_{\parallel}(d,b,\delta) = -\frac{\kappa_d}{2}\delta + O(|b|\delta + \delta^2), \tag{27}$$

$$\lambda_{\perp}(d,b,\delta) = b^2\kappa_d^2 + O(|b|^3 + \delta). \tag{28}$$

All remainders are uniform for the indicated parameters.

Proof. Write $\Phi = \Phi_{d,b}$, set $t = w + \delta\mathbf{1}/2$, and put $f(s) = \Phi(w + s\mathbf{1})$. Centered Taylor expansion gives

$$f(-\delta/2) - f(\delta/2) = -\delta f'(0) - \frac{\delta^3}{24}f'''(0) + O(\delta^5).$$

Since $f^{(j)}(0) = \partial_{\mathbf{1}}^j\Phi(w)$, division by δ yields, in every fixed C^r norm,

$$\frac{R_{d,b,\delta}(w + \frac{\delta}{2}\mathbf{1})}{\delta} = -\partial_{\mathbf{1}}\Phi_{d,b}(w) - \frac{\delta^2}{24}\partial_{\mathbf{1}}^3\Phi_{d,b}(w) + O(\delta^4). \tag{29}$$

Denote the left side by $Q_\delta(w)$, with $Q_0 = S_{d,b}$. The C^2 estimate $Q_\delta - Q_0 = O(\delta^2)$ preserves the strict concavity and value gap from Proposition 2.3. Thus Q_δ has a unique global maximum w_δ near $w_0 = \sigma_{d,b}\mathbf{1}$.

Writing $\mathbf{h} = w_\delta - w_0$ and using stationarity,

$$\left(\int_0^1 \nabla_t^2 S_{d,b}(w_0 + s\mathbf{h})\, ds\right) \mathbf{h} = \nabla S_{d,b}(w_\delta) - \nabla Q_\delta(w_\delta) = O(\delta^2).$$

The matrix has a uniformly bounded inverse, so $\mathbf{h} = O(\delta^2)$.

Exchange symmetry gives $w_\delta = (s_\delta, s_\delta)$. Hence

$$t_\delta = w_\delta + \frac{\delta}{2}\mathbf{1} = \left(\sigma_{d,b} + \frac{\delta}{2} + O(\delta^2)\right)\mathbf{1}.$$

This proves (26). The eigenvectors remain $(1,1)$ and $(1,-1)$ along the diagonal. With $\sigma = \sigma_{d,b}$ and $v = v_{d,b,\delta}$, Taylor expansion gives

$$\begin{aligned}\lambda_\parallel(d,b,\delta) &= \Lambda_\parallel(\sigma) + \Lambda_\parallel'(\sigma)(v-\sigma) + O((v-\sigma)^2)\\ &= 0 + (-\kappa_d + O(b))\left(\frac{\delta}{2} + O(\delta^2)\right) + O(\delta^2)\\ &= -\frac{\kappa_d}{2}\delta + O(|b|\delta + \delta^2).\end{aligned}$$

For the other fixed direction, put $\Lambda_\perp(s) = -(\Phi_{11} - \Phi_{12})(s,s)$. Its derivative is uniformly bounded, and hence

$$\Lambda_\perp(v) = \Lambda_\perp(\sigma) + O(|v-\sigma|) = b^2\kappa_d^2 + O(|b|^3) + O(\delta).$$

These are (27) and (28). □

Remark 2.5. The exact identity (23) excludes an error term independent of δ in (27), which is needed for the sign when $\delta = o(b^2)$.

Figure 2 summarizes the local sign change. Here $\Lambda_\perp(t)$ denotes the transverse eigenvalue of $\mathcal{H}_{d,b}(it, it)$, alongside $\Lambda_\parallel(t)$.

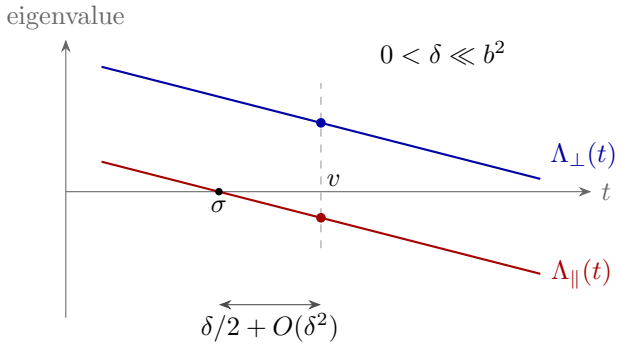


FIGURE 2. Local eigenvalue behavior for fixed small $b > 0$, with $\sigma = \sigma_{d,b}$ and $v = v_{d,b,\delta}$. The longitudinal eigenvalue vanishes at σ and becomes negative at v, while the transverse eigenvalue remains positive. Straight segments indicate the local behavior only.

3. Period normalization from harmonic measures

We fix the orientations and constants relating the period matrix to harmonic measures. Let ϕ denote the canonical antiholomorphic involution of the Schottky double, exchanging the two copies of Ω and fixing their common boundary pointwise.

Let Γ_0 be the outer boundary of a regular $(g+1)$-connected planar domain, and let $\Gamma_1, \dots, \Gamma_g$ be the inner boundaries. Let ω_j be the harmonic function with boundary values 1 on Γ_j and 0 on the other components. Define

$$C_{jk} = \int_\Omega \nabla\omega_j \cdot \nabla\omega_k \, dA = \int_{\Gamma_j} \partial_\nu \omega_k \, ds. \tag{30}$$

Here ν is the outward unit normal of Ω, pointing into each hole. Green's first identity gives

$$\begin{aligned}\int_\Omega \nabla\omega_j \cdot \nabla\omega_k \, dA &= \int_{\partial\Omega} \omega_j \partial_\nu \omega_k \, ds - \int_\Omega \omega_j \Delta\omega_k \, dA \\ &= \sum_{\ell=0}^{g} \int_{\Gamma_\ell} \omega_j \partial_\nu \omega_k \, ds = \int_{\Gamma_j} \partial_\nu \omega_k \, ds.\end{aligned}$$

C is symmetric. For $\mathbf{x} \in \mathbb{R}^g$ and $v = \sum_j x_j \omega_j$,

$$\mathbf{x}^T C \mathbf{x} = \int_\Omega \left| \sum_j x_j \nabla\omega_j \right|^2 dA = \int_\Omega |\nabla v|^2 dA \geq 0.$$

Equality forces v to be constant. Its boundary values $v|_{\Gamma_0} = 0$ and $v|_{\Gamma_j} = x_j$ then give $\mathbf{x} = 0$, proving $C > 0$. We call $\int_\Gamma \partial_\nu v \, ds$ the *normal flux* of v through Γ.

Use the Hodge star $*dx = dy$, $*dy = -dx$, and set

$$\eta_k = -i\pi(d\omega_k + i * d\omega_k). \tag{31}$$

In local coordinates $z = x + iy$,

$$d\omega_k + i * d\omega_k = ((\omega_k)_x - i(\omega_k)_y)(dx + i\, dy) = 2\partial\omega_k.$$

Harmonicity makes this form holomorphic. Since $d\omega_k$ vanishes on boundary tangent vectors, $\eta_k = \pi * d\omega_k - i\pi d\omega_k$ is real there and extends to the double by Schwarz reflection, with $\phi^*\eta_k = \overline{\eta_k}$.

Orient B_j counterclockwise around the jth hole. Choose disjoint crosscuts from Γ_j to Γ_0 in the front copy and return along their reflections to form the cycles A_j. These orientations give a canonical basis. Write $\gamma_j : [0,1] \to \overline{\Omega}$ for the front crosscut, with $\gamma_j(0) \in \Gamma_j$ and $\gamma_j(1) \in \Gamma_0$, and put $\mathcal{J}_{jk} = \int_{\gamma_j} \eta_k$. Since $\eta_k = \pi * d\omega_k - i\pi d\omega_k$,

$$\operatorname{Im} \mathcal{J}_{jk} = -\pi \int_{\gamma_j} d\omega_k = -\pi\big(\omega_k(\gamma_j(1)) - \omega_k(\gamma_j(0))\big) = -\pi(0 - \delta_{jk}) = \pi\delta_{jk},$$

where δ_{jk} is the Kronecker delta. The reflected return path contributes $-\overline{\mathcal{J}_{jk}}$, so the full A_j-period is $\mathcal{J}_{jk} - \overline{\mathcal{J}_{jk}} = 2i \operatorname{Im} \mathcal{J}_{jk}$. This gives

$$\int_{A_j} \eta_k = 2\pi i \delta_{jk}. \tag{32}$$

For the counterclockwise unit tangent $T = (T_x, T_y)$ to B_j, the right normal $(T_y, -T_x)$ equals $-\nu$. Thus

$$(*d\omega_k)(T) = (\omega_k)_x T_y - (\omega_k)_y T_x = \nabla \omega_k \cdot (T_y, -T_x) = -\partial_\nu \omega_k.$$

Since $\int_{B_j} d\omega_k = 0$, integration gives

$$\tau_{jk} = \int_{B_j} \eta_k = -\pi C_{jk}. \tag{33}$$

The matrix τ is real symmetric and negative definite. The orientations are shown in Figure 3.

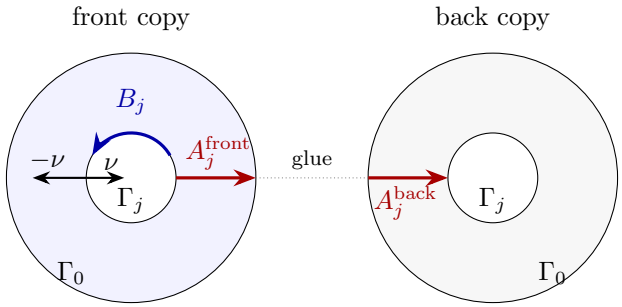


FIGURE 3. A schematic for one inner boundary component; other holes are omitted. Corresponding inner and outer boundaries are identified in the double. The cycle A_j goes from Γ_j to Γ_0 in front and returns in back. In front, B_j is counterclockwise. Its flux normal $-\nu$ points out of the hole, whereas the outward normal ν of the domain points into the hole.

Lemma 3.1. *Evaluate $p - \phi(p)$ along a symmetric path from $\phi(p)$ to p passing through Γ_0. Then, in the above normalized coordinates,*

$$p - \phi(p) = -2\pi i\big(\omega_1(p), \dots, \omega_g(p)\big). \tag{34}$$

Proof. Let J_k be the integral of η_k on the front half of the path, from Γ_0 to p. The full integral is $J_k - \overline{J_k}$, and $\operatorname{Im} J_k = -\pi \omega_k(p)$. This is (34). Changing the admissible paths by A-periods only adds $2\pi i \mathbb{Z}^g$, which does not affect theta values or their Hessians. □

Remark 3.2 (An annulus check). For the annulus $\rho < |z| < 1$, the harmonic measure is $\log|z| / \log \rho$ and its energy is $2\pi / \log(1/\rho)$. Formula (33) gives the scalar period $-2\pi^2 / \log(1/\rho)$. This verifies the same factor d_* as in (6).

4. Explicit estimates for the circle domains

Fix ρ and let $c \to 1^-$ in the domains (8). Reflection $z \mapsto -\overline{z}$ interchanges the holes, so uniqueness of the Dirichlet problem gives $\omega_+(-\overline{z}) = \omega_-(z)$. Consequently,

$$\pi C = \begin{pmatrix} d & -b \\ -b & d \end{pmatrix}. \tag{35}$$

where

$$d := \pi C_{++} = \pi C_{--}, \qquad b := -\pi C_{+-}, \qquad \tau = -\pi C = \begin{pmatrix} -d & b \\ b & -d \end{pmatrix}.$$

The Hopf lemma gives $C_{+-} < 0$, since ω_+ is positive inside Ω_c and vanishes on ∂K_-. Thus $b > 0$, and $C > 0$ implies $d > b$.

4.1. **Comparison with the exact single-hole solutions.**

Put

$$W_\pm(z) = -\frac{1}{M_\rho} \log \left| \frac{z \mp c}{1 \mp cz} \right|. \tag{36}$$

These are the single-hole harmonic measures, with flux

$$C_* = \frac{2\pi}{M_\rho}, \qquad d_* = \pi C_*. \tag{37}$$

$w = (z-c)/(1-cz)$ maps the one-hole domain to $\rho < |w| < 1$, where $W_+ = -\log|w|/M_\rho$. On $|w| = \rho$, the outward normal is $-\partial_r$, so

$$\partial_{\nu_w} \left(-\frac{\log r}{M_\rho} \right) = \frac{1}{\rho M_\rho}, \qquad \int_{|w|=\rho} \frac{1}{\rho M_\rho}\, ds_w = \int_0^{2\pi} \frac{1}{\rho M_\rho} \rho\, d\theta = \frac{2\pi}{M_\rho}.$$

Flux is conformally invariant: for $w = T(z)$ and $v = \widetilde{v} \circ T$,

$$\partial_{\nu_z} v = |T'(z)| \partial_{\nu_w} \widetilde{v}, \qquad ds_w = |T'(z)| ds_z, \qquad \partial_{\nu_z} v\, ds_z = \partial_{\nu_w} \widetilde{v}\, ds_w.$$

The same calculation applies to W_-.

For the interaction parameter define

$$a = \frac{2c}{1+c^2}, \qquad \gamma = -\log a. \tag{38}$$

For c sufficiently close to 1, $a > \rho$. The automorphism $u = (z+c)/(1+cz)$ maps K_- to $\{|u| \le \rho\}$, and in this coordinate

$$W_+(z(u)) = -\frac{1}{M_\rho} \log \left| \frac{u-a}{1-au} \right|. \tag{39}$$

$z = (u-c)/(1-cu)$ gives

$$\frac{z-c}{1-cz} = \frac{(1+c^2)u - 2c}{1+c^2-2cu} = \frac{u-a}{1-au}.$$

On $u = \rho e^{i\theta}$, its squared modulus is, with $x = \cos\theta$,

$$\mathcal{R}(x) = \frac{a^2+\rho^2-2a\rho x}{1+a^2\rho^2-2a\rho x}, \qquad \mathcal{R}'(x) = -\frac{2a\rho(1-a^2)(1-\rho^2)}{(1+a^2\rho^2-2a\rho x)^2} < 0.$$

The minimum and maximum of W_+ on ∂K_- occur at $x = -1$ and $x = 1$, respectively, and equal

$$\alpha = \frac{1}{M_\rho} \log \frac{1 + a\rho}{a + \rho}, \qquad \beta = \frac{1}{M_\rho} \log \frac{1 - a\rho}{a - \rho}. \tag{40}$$

Both are positive and unchanged when the holes are interchanged. Since

$$\begin{aligned} \alpha(1) &= \beta(1) = 0, \\ \alpha'(1) &= \frac{1}{M_\rho} \left(\frac{\rho}{1+\rho} - \frac{1}{1+\rho} \right) = -\frac{1-\rho}{M_\rho(1+\rho)}, \\ \beta'(1) &= \frac{1}{M_\rho} \left(-\frac{\rho}{1-\rho} - \frac{1}{1-\rho} \right) = -\frac{1+\rho}{M_\rho(1-\rho)}, \end{aligned}$$

and $a = e^{-\gamma} = 1 - \gamma + O(\gamma^2)$, Taylor expansion gives

$$\alpha = \frac{1-\rho}{M_\rho(1+\rho)}\gamma + O(\gamma^2), \qquad \beta = \frac{1+\rho}{M_\rho(1-\rho)}\gamma + O(\gamma^2). \tag{41}$$

Lemma 4.1. *Let $V_+ = W_+ - \omega_+$. Then*

$$\alpha\omega_- \le V_+ \le \beta\omega_- \quad \text{on } \Omega_c. \tag{42}$$

The analogous inequalities hold with the labels interchanged. Moreover,

$$\alpha d \le b \le \beta d, \tag{43}$$

$$\alpha b \le d - d_* \le \beta b. \tag{44}$$

In particular, when $\beta < 1$,

$$\frac{d_*}{1-\alpha^2} \le d \le \frac{d_*}{1-\beta^2}. \tag{45}$$

Thus $d = d_* + O(\gamma^2)$ and $b \asymp \gamma$.

Proof. The function V_+ is harmonic, vanishes on Γ_0 and ∂K_+, and equals $W_+ \in [\alpha, \beta]$ on ∂K_-. The maximum principle proves (42).

Green's second identity gives

$$0 = \int_{\Omega_c} (\omega_- \Delta V_+ - V_+ \Delta\omega_-)\, dA = \int_{\partial\Omega_c} (\omega_- \partial_\nu V_+ - V_+ \partial_\nu \omega_-)\, ds.$$

Only ∂K_- contributes; there $\omega_- = 1$ and $V_+ = W_+$. Hence

$$\int_{\partial K_-} \partial_\nu V_+\, ds = \int_{\partial K_-} W_+ \partial_\nu \omega_-\, ds. \tag{46}$$

Since W_+ is harmonic across K_-, the divergence theorem gives

$$\int_{\partial K_-} \partial_\nu W_+\, ds = -\int_{K_-} \Delta W_+\, dA = 0.$$

Therefore

$$\int_{\partial K_-} \partial_\nu V_+\, ds = \int_{\partial K_-} (\partial_\nu W_+ - \partial_\nu \omega_+)\, ds = 0 - C_{-+} = \frac{b}{\pi}.$$

Since $\partial_\nu \omega_- > 0$ on ∂K_- and its integral is $C_{--} = d/\pi$, (46) gives (43).

On ∂K_+, all terms in (42) vanish. Taking outward normal derivatives reverses the inequalities, since a nonnegative function vanishing at the boundary has nonpositive outward derivative:

$$\beta\partial_\nu\omega_- \le \partial_\nu V_+ \le \alpha\partial_\nu\omega_-.$$

The integral of the middle term is

$$\int_{\partial K_+} \partial_\nu V_+ \, ds = \int_{\partial K_+} \partial_\nu W_+ \, ds - \int_{\partial K_+} \partial_\nu \omega_+ \, ds = C_* - C_{++}.$$

Therefore

$$\beta C_{+-} \le C_* - C_{++} \le \alpha C_{+-}.$$

Substituting $C_{+-} = -b/\pi$ and $C_* - C_{++} = (d_* - d)/\pi$ proves (44). Together with (43), this gives $\alpha^2 d \le d - d_* \le \beta^2 d$ and hence (45). The asymptotics follow from (41). □

4.2. **The leading interaction coefficient.** We now determine the leading coefficient of b.

Proposition 4.2. *As $c \to 1^-$,*

$$b = \frac{2\pi^2}{M_\rho^2}\gamma + O(\gamma^3). \tag{47}$$

With $c = 1 - \varepsilon$,

$$d = d_* + O(\varepsilon^4), \qquad b = \frac{\pi^2}{M_\rho^2}\varepsilon^2 + O(\varepsilon^3). \tag{48}$$

Proof. In the coordinate (39), fix $\rho < r_0 < 1$. For c near 1, the other hole lies outside $|u| \le r_0$. The cross potential W_+ is $O(\gamma)$ there, with any fixed number of derivatives, because it depends smoothly on a and vanishes at $a = 1$.

Use (42) with the labels interchanged:

$$0 \le V_- := W_- - \omega_- \le \beta\omega_+ \le \beta W_+.$$

Thus $V_- = O(\gamma^2)$ on the fixed collar $\rho \le |u| \le r_0$, and $V_- = 0$ on $|u| = \rho$. Choose M independent of c so that $V_- \le M\gamma^2$ on $|u| = r_0$. The radial harmonic function

$$B(u) = M\gamma^2 \frac{\log(|u|/\rho)}{\log(r_0/\rho)}$$

has the required boundary values, so $0 \le V_- \le B$. On $|u| = \rho$, where $\nu_u = -\partial_r$,

$$-\frac{M\gamma^2}{\rho \log(r_0/\rho)} = \partial_{\nu_u} B \le \partial_{\nu_u} V_- \le 0.$$

Since $W_-(z(u)) = -\log|u|/M_\rho$, the transformed harmonic measure $\widetilde{\omega}_- = \omega_- \circ z$ satisfies

$$\partial_{\nu_u}\widetilde{\omega}_-(u) = \frac{1}{\rho M_\rho} + O(\gamma^2), \qquad |u| = \rho. \tag{49}$$

Flux measure is conformally invariant, so (46) becomes

$$\frac{b}{\pi} = \int_{|u|=\rho} \left(-\frac{1}{M_\rho} \log \left| \frac{u-a}{1-au} \right| \right) \partial_{\nu_u} \widetilde{\omega}_-(u)\, ds_u.$$

Denote the logarithmic factor by $\mathcal{W}$ and the error in (49) by E. Since $\mathcal{W}$ is harmonic on $|u| \le \rho$, $\mathcal{W}(0) = \gamma/M_\rho$, $\mathcal{W} = O(\gamma)$ and $E = O(\gamma^2)$,

$$\int_{|u|=\rho} \mathcal{W}(u)\, ds_u = 2\pi\rho\mathcal{W}(0) = \frac{2\pi\rho\gamma}{M_\rho}, \qquad \left| \int_{|u|=\rho} \mathcal{W}(u) E(u)\, ds_u \right| = O(\gamma^3).$$

Inserting (49) therefore gives

$$\frac{b}{\pi} = \frac{1}{\rho M_\rho} \frac{2\pi\rho\gamma}{M_\rho} + O(\gamma^3),$$

which proves (47). Lastly,

$$\gamma = \log \frac{1+c^2}{2c} = \log \left(1 + \frac{\varepsilon^2}{2(1-\varepsilon)} \right) = \frac{1}{2}\varepsilon^2 + O(\varepsilon^3).$$

This and Lemma 4.1 prove (48). □

4.3. **A pole whose displacement is smaller than the squared coupling.**

Proposition 4.3. *At $p_\varepsilon = i(1-\varepsilon^5)$,*

$$\omega_+(p_\varepsilon) = \omega_-(p_\varepsilon) =: \mu_\varepsilon, \tag{50}$$

and

$$\delta_\varepsilon := 2\pi\mu_\varepsilon = \frac{2\pi}{M_\rho}\varepsilon^6 + O(\varepsilon^7). \tag{51}$$

In particular $\delta_\varepsilon / b_\varepsilon^2 \to 0$.

Proof. The reflection $z \mapsto -\overline{z}$ fixes p_ε, so the two measures are equal. The single-hole potentials are also equal there; denote their common value by w_ε. Evaluating (42) at this point gives

$$\alpha\mu_\varepsilon \le w_\varepsilon - \mu_\varepsilon \le \beta\mu_\varepsilon, \qquad \frac{w_\varepsilon}{1+\beta} \le \mu_\varepsilon \le \frac{w_\varepsilon}{1+\alpha}.$$

Since $\alpha, \beta = O(\varepsilon^2)$, it follows that $\mu_\varepsilon = w_\varepsilon(1 + O(\varepsilon^2))$.

For $y = 1-\varepsilon^5$ and $c = 1-\varepsilon$, the value is explicit:

$$\begin{aligned} w_\varepsilon &= -\frac{1}{2M_\rho} \log \frac{y^2+c^2}{1+c^2y^2} \\ &= -\frac{1}{2M_\rho} \log \left(1 - \frac{(1-c^2)(1-y^2)}{1+c^2y^2} \right). \end{aligned} \tag{52}$$

Here

$$1-c^2 = 2\varepsilon - \varepsilon^2, \qquad 1-y^2 = 2\varepsilon^5 - \varepsilon^{10},$$
$$(1-c^2)(1-y^2) = 4\varepsilon^6 + O(\varepsilon^7), \qquad 1+c^2y^2 = 2 + O(\varepsilon).$$

The fraction is $2\varepsilon^6 + O(\varepsilon^7)$. Since $-\log(1-x) = x + O(x^2)$,

$$w_\varepsilon = \frac{1}{2M_\rho}\big(2\varepsilon^6 + O(\varepsilon^7) + O(\varepsilon^{12})\big) = \frac{\varepsilon^6}{M_\rho} + O(\varepsilon^7).$$

It follows that $\mu_\varepsilon = \varepsilon^6/M_\rho + O(\varepsilon^7)$, proving (51). Proposition 4.2 gives

$$b_\varepsilon^2 = \frac{\pi^4}{M_\rho^4}\varepsilon^4 + O(\varepsilon^5), \qquad \frac{\delta_\varepsilon}{b_\varepsilon^2} = \frac{2M_\rho^3}{\pi^3}\varepsilon^2 + O(\varepsilon^3) \longrightarrow 0.$$

□

Remark 4.4 (Choice of the pole scale). The exponent 5 in (8) is not essential. If $p_{\varepsilon,m} = i(1-\varepsilon^m)$ with a fixed $m > 3$, the same calculation in (52) gives

$$\omega_+(p_{\varepsilon,m}) = \frac{\varepsilon^{m+1}}{M_\rho}(1 + O(\varepsilon)), \qquad \delta_{\varepsilon,m} = \frac{2\pi}{M_\rho}\varepsilon^{m+1}(1 + O(\varepsilon)).$$

Since $b_\varepsilon^2 \asymp \varepsilon^4$, one has $\delta_{\varepsilon,m}/b_\varepsilon^2 \asymp \varepsilon^{m-3} \to 0$. Proposition 2.4 then gives a negative longitudinal eigenvalue of order ε^{m+1} and a positive transverse eigenvalue of order ε^4. In particular, $m = 4$ suffices; $m = 5$ is a convenient fixed choice.

5. Identification with the Green-function parameter

We use Yamada's extremal characterization of $e_0(p)$.

Theorem 5.1 (Yamada, [Yam99, Theorem 2.1])**.** *Let Ω be a regular planar domain, and use the normalized symmetric homology basis and theta convention of Section 3. For $p \in \Omega$, evaluate $p - \phi(p)$ along a symmetric path. On $i\mathbb{R}^g/(2\pi i\mathbb{Z}^g)$ the positive function*

$$e \longmapsto \frac{\theta_\tau(p - \phi(p) + e)}{\theta_\tau(e)} \tag{53}$$

has its unique maximum at $e = e_0(p)$, defined by (4).

The conventions agree with [Yam99, pp. 1400–1402]: the boundary cycles are the B-cycles, the A-periods are $2\pi i$, the theta exponent is $\frac{1}{2}n^T\tau n + n^T e$, and $p - \phi(p)$ is integrated along a symmetric path. Yamada uses the positive Green function $g = -G_\Omega$; changing its sign leaves the critical divisor D_p unchanged. His upper equality case is therefore precisely the parameter in (4). The zeros in D_p are counted with their orders of vanishing.

Remark 5.2 (The genus-one case). For an annulus, put $\delta = 2\pi\omega_1(p) \in (0, 2\pi)$. The logarithm of the ratio in (53) is

$$r_\delta(t) = \ell_d(t - \delta) - \ell_d(t).$$

At a maximum t_*, stationarity gives $\ell_d'(t_* - \delta) = \ell_d'(t_*)$. The proof of Lemma 2.1 shows that $\ell_d'' < 0$ on $(-t_d, t_d)$ and $\ell_d'' > 0$ on $(t_d, 2\pi - t_d)$, modulo 2π. On each arc ℓ_d' is strictly monotone, and its endpoint extrema

are attained only once. Since $\delta \notin 2\pi\mathbb{Z}$, the two arguments must lie in different open arcs. The maximum condition

$$r''_\delta(t_*) = \ell''_d(t_* - \delta) - \ell''_d(t_*) \le 0$$

forces $\ell''_d(t_*) > 0$. Thus $H_{-d}(it_*) = -\ell''_d(t_*) < 0$, in agreement with Saitoh's doubly connected result.

Proof of Theorem 1.1. For Ω_ε, equations (33) and (35) identify its theta function with $\Theta_{d_\varepsilon,b_\varepsilon}$. The parameters satisfy

$$d_\varepsilon = d_* + O(\varepsilon^4), \qquad b_\varepsilon = \frac{\pi^2}{M_\rho^2}\varepsilon^2 + O(\varepsilon^3) > 0.$$

By Lemma 3.1 and Proposition 4.3,

$$p_\varepsilon - \phi(p_\varepsilon) = -i\delta_\varepsilon(1,1), \qquad \delta_\varepsilon = \frac{2\pi}{M_\rho}\varepsilon^6 + O(\varepsilon^7) > 0.$$

For $e = it$, the logarithm of (53) is $R_{d_\varepsilon,b_\varepsilon,\delta_\varepsilon}(t)$. Uniqueness in Theorem 5.1 and Proposition 2.4 therefore gives

$$e_0(p_\varepsilon) = iv_{d_\varepsilon,b_\varepsilon,\delta_\varepsilon}(1,1) \pmod{2\pi i\mathbb{Z}^2}. \tag{54}$$

Smooth dependence in Lemma 2.1 yields $\kappa_{d_\varepsilon} = \kappa_* + O(\varepsilon^4)$. Substitution into (27) gives

$$\lambda_\parallel = -\frac{\kappa_*}{2}\frac{2\pi}{M_\rho}\varepsilon^6 + O(\varepsilon^7) + O(\varepsilon^8) + O(\varepsilon^{12}),$$

which proves (12). Since $|b_\varepsilon|^3 + \delta_\varepsilon = O(\varepsilon^6)$, (28) gives

$$\lambda_\perp = \kappa_*^2\left(\frac{\pi^2}{M_\rho^2}\varepsilon^2 + O(\varepsilon^3)\right)^2 + O(\varepsilon^6) = \frac{\pi^4\kappa_*^2}{M_\rho^4}\varepsilon^4 + O(\varepsilon^5).$$

This proves (13); the signs follow from $M_\rho, \kappa_* > 0$. □

Corollary 5.3. *For c sufficiently close to 1, fix the domain Ω_c with the two holes (7). There exists $y_0(c) < 1$ such that $H_{\tau_c}(e_0(iy))$ is indefinite for every $y_0(c) < y < 1$.*

Proof. Choose c close enough to 1 that $d_c \in \mathcal{I}$ and $b_c > 0$ is small. Proposition 2.3 gives a positive transverse eigenvalue at $\delta = 0$, while (27) gives

$$\lambda_\parallel(d_c, b_c, \delta) = \delta\left(-\frac{\kappa_{d_c}}{2} + O(b_c) + O(\delta)\right).$$

Choose c so that $|O(b_c)| < \kappa_{d_c}/4$ and then fix it. For sufficiently small $\delta > 0$, the longitudinal eigenvalue is negative and the transverse one remains positive. For y sufficiently close to 1, the point iy lies in Ω_c. Reflection symmetry gives $\omega_+(iy) = \omega_-(iy)$; their common value is positive and tends to zero as $y \uparrow 1$. Set $\delta(y) = 2\pi\omega_+(iy)$. Lemma 3.1 gives

$$iy - \phi(iy) = -i\delta(y)\mathbf{1}.$$

The logarithm of Yamada's ratio at this pole is exactly $R_{d_c,b_c,\delta(y)}$. Proposition 2.4 places its unique maximizer at $t = v_{d_c,b_c,\delta(y)}\mathbf{1}$. Passing to $e = it$,

Theorem 5.1 identifies $iv_{d_c,b_c,\delta(y)}\mathbf{1}$ with $e_0(iy)$ modulo $2\pi i\mathbb{Z}^2$. Choose $y_0(c)$ so that $0<\delta(y)$ stays within the preceding sign bounds whenever $y_0(c)<y<1$. The Hessian at $e_0(iy)$ then has the two asserted opposite signs. □

6. Kernel comparison, directional negativity, and higher genus

Fix $p\in\Omega$. Let $G_\Omega(\cdot,p)$ be the Green function with singularity $\log|z-p|$, and put $\varpi_p=\partial_\nu G_\Omega(\cdot,p)>0$ on $\partial\Omega$. Denote by R_p and $\widehat{R}_p$ the reproducing kernels for the boundary norms

$$\|f\|_{p,+}^2=\frac{1}{2\pi}\int_{\partial\Omega}|f|^2\varpi_p\,ds,\qquad \|f\|_{p,-}^2=\frac{1}{2\pi}\int_{\partial\Omega}|f|^2\varpi_p^{-1}\,ds,$$

respectively, and by K_Ω the Bergman kernel for the area norm $\int_\Omega|f|^2dA$. These are the Hardy and conjugate Hardy kernels in [Yam99, Lemma 3.1 and Remark 3.1], expressed in the planar coordinate. Saitoh's kernel-order question [Sai79, pp. 276–277, equation (7.11)] asks whether

$$\mathscr{K}_p(z,\overline{w}):=R_p(z,\overline{w})\widehat{R}_p(z,\overline{w})-\pi K_\Omega(z,\overline{w})\gg 0. \tag{55}$$

Here $\gg 0$ means that every finite matrix $\big(\mathscr{K}_p(z_\ell,\overline{z_m})\big)_{\ell,m}$ is positive semi-definite; it is stronger than a pointwise diagonal inequality.

Write $\eta_j=f_j(z)\,dz$ for the normalized differentials of Section 3, and set $U(z)=(f_1(z),\dots,f_g(z))^T$. Multiplying the theta formulas for R_p and $\widehat{R}_p$ and applying Fay's identity gives

$$\mathscr{K}_p(z,\overline{w})=-U(z)^TH_\tau(e_0(p))\overline{U(w)}; \tag{56}$$

see [Yam99, Lemma 3.1] and [Yam98, pp. 40–41, equations (5), (6), (10)]. Since the f_j are linearly independent, the vectors $U(z)$ span $\mathbb{C}^g$. For points $z_1,\dots,z_N$ and coefficients $c_1,\dots,c_N$, set $\xi=\sum_{\ell=1}^N c_\ell\overline{U(z_\ell)}$. Formula (56) gives

$$\sum_{\ell,m=1}^N\overline{c_\ell}\,\mathscr{K}_p(z_\ell,\overline{z_m})c_m=-\xi^*H_\tau(e_0(p))\xi.$$

The conjugate on $U(w)$ in (56) makes the kernel antiholomorphic in w. The spanning property therefore gives the exact equivalence

$$\mathscr{K}_p\gg 0\quad\Longleftrightarrow\quad H_\tau(e_0(p))\le 0. \tag{57}$$

Strict negative definiteness in (5) is equivalent to (55) and the rank condition $\operatorname{rank}\mathscr{K}_p=g$. Here the rank of a kernel is the supremum of the ranks of its finite Gram matrices. Our indefinite examples violate even (55).

Since $H_\tau(e_0(p))$ is real symmetric,

$$U(p)^TH_\tau(e_0(p))\overline{U(p)}=U(p)^*H_\tau(e_0(p))U(p).$$

Harmonic measure gives $R_p(z,\overline{p})=1$. Setting $z=w=p$ in (56) gives

$$\widehat{R}_p(p,\overline{p})-\pi K_\Omega(p,\overline{p})=-U(p)^*H_\tau(e_0(p))U(p)>0, \tag{58}$$

where strict positivity for $g \geq 1$ is Guan's theorem [Gua19, Theorem 1.1]. The distinguished direction is explicitly

$$U(p) = -2\pi i \left(\frac{\partial \omega_1}{\partial z}(p), \dots, \frac{\partial \omega_g}{\partial z}(p) \right)^T. \tag{59}$$

Guan's result gives strict negativity on the complex line $\mathbb{C}U(p)$. For $g = 1$ this line is the whole space; for $g \geq 2$ it is a proper subspace, which is compatible with an indefinite Hessian.

The construction also extends to every $g \geq 2$. Fix a triply connected example $(\Omega^{(2)}, p)$ from Theorem 1.1. For $g > 2$, choose distinct $\zeta_1, \dots, \zeta_{g-2} \in \partial\mathbb{D}$ and $0 < r < 1$, and remove the additional disks

$$J_{j,\chi} = \left\{ z \in \mathbb{D} : \left| \frac{z - a_{j,\chi}}{1 - \overline{a_{j,\chi}} z} \right| \leq r \right\}, \qquad a_{j,\chi} = (1 - \chi)\zeta_j.$$

For small $\chi > 0$, these disks are mutually disjoint and avoid the old holes and p. Let $\Omega_\chi = \Omega^{(2)} \setminus \bigcup_j J_{j,\chi}$ and $d_r = 2\pi^2 / \log(1/r)$. The single-hole potentials $-\log|(z - a_{j,\chi})/(1 - \overline{a_{j,\chi}} z)| / \log(1/r)$ tend to zero on compact subsets of $\Omega^{(2)}$. The comparison and flux arguments of Section 4 give, with the old cycles listed first,

$$\tau_\chi \longrightarrow \operatorname{diag}(\tau^{(2)}, -d_r I_{g-2}), \qquad \omega_\chi(p) \longrightarrow (\omega^{(2)}(p), 0). \tag{60}$$

The new single-hole potentials are $O(\chi)$ on the old boundaries and $O(\chi^2)$ on the other new holes. The maximum principle and boundary estimates give convergence of the old harmonic measures and their fluxes, and vanishing mixed fluxes. Each new diagonal capacity is at least the single-hole capacity; cutting off its single-hole potential in a fixed neighborhood of ζ_j gives the reverse bound up to $O(\chi^2)$. Thus its limit is $2\pi / \log(1/r)$. The cutoffs may be chosen with disjoint supports; Dirichlet minimality then makes the difference between their diagonal energy matrix and the new capacity block positive semidefinite. Its diagonal entries tend to zero, so the off-diagonal entries do as well. The theta functions consequently converge in C^2 on the imaginary torus to $\theta_{\tau^{(2)}}(it_1, it_2) \prod_{j=3}^{g} P_{d_r}(t_j)$. The limiting theta ratio is independent of $t_3, \dots, t_g$ and is maximal precisely when $i(t_1, t_2) = e_0^{(2)}(p)$ modulo periods. Uniform convergence, Theorem 5.1, and compactness force the first two coordinates of $e_{0,\chi}(p)$ to converge to this point. Hence the upper-left 2×2 block of $H_{\tau_\chi}(e_{0,\chi}(p))$ converges to the original indefinite Hessian. Its two strict signs persist, proving indefiniteness on these $(g+1)$-connected domains for all sufficiently small $\chi > 0$.

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Declaration on the use of artificial intelligence

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